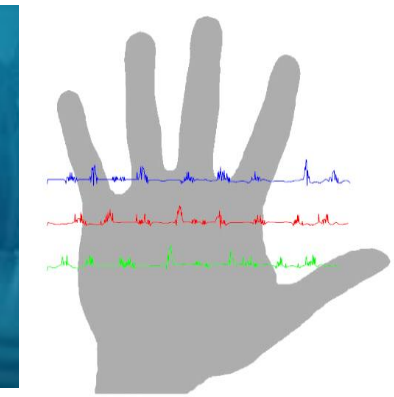
PalmPrint Identification System

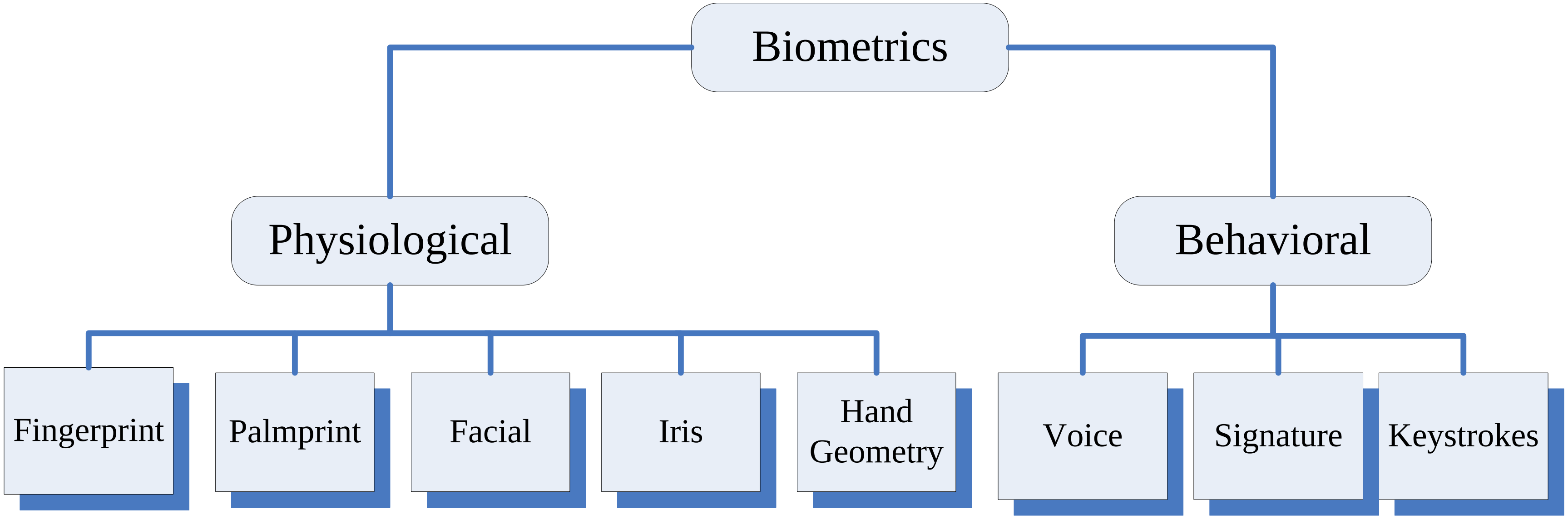


CONTENTS

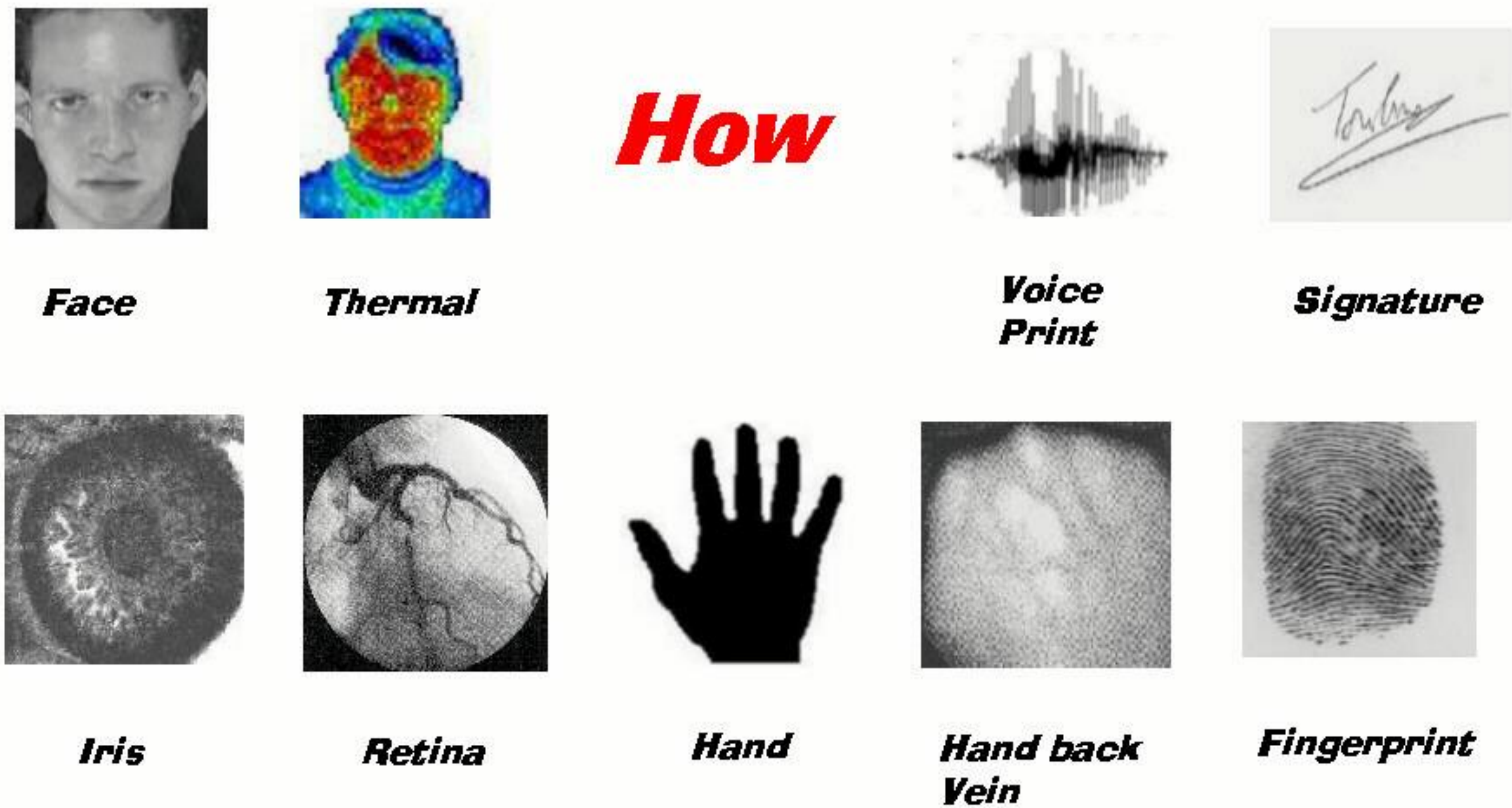
Biometrics
Which Biometric is the Best?
Recognition Flow Chart
Feature Extraction
Experiment & Results
Conclusion

Biometric Recognition System

- **Biometrics** :is the automated use of physiological or behavioral characteristics to determine or verify and identity a person

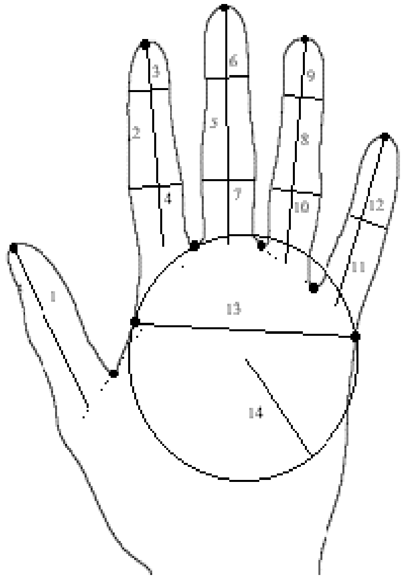


Biometrics Examples

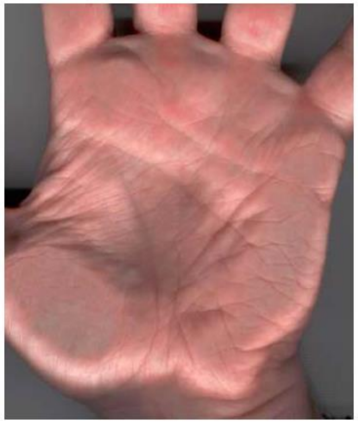


Hand and Palm Recognition

- **Features: dimensions and shape of the hand, fingers, (size and length)**



- **Features: Palmprint focuses on the inner surface of a hand, its pattern of lines and the shape of its surface.**



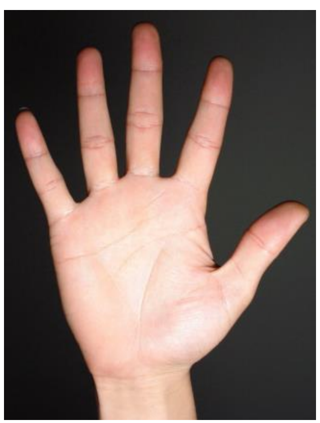
Which Biometric is the Best?

Why PalmPrint?

- High Distinctiveness
- High Permanence (duration)
- High Performance
- Medium Collectabillity
- Medium Acceptability
- Medium Universality
- Medium Circumvention (fooling)

Palmprint Recognition Flowchart

Image acquisition → Image preprocessing → Feature extraction → Classification



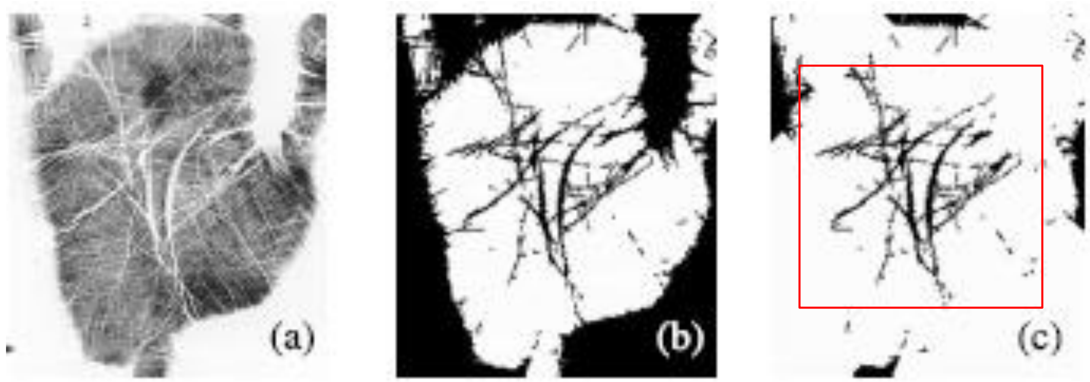
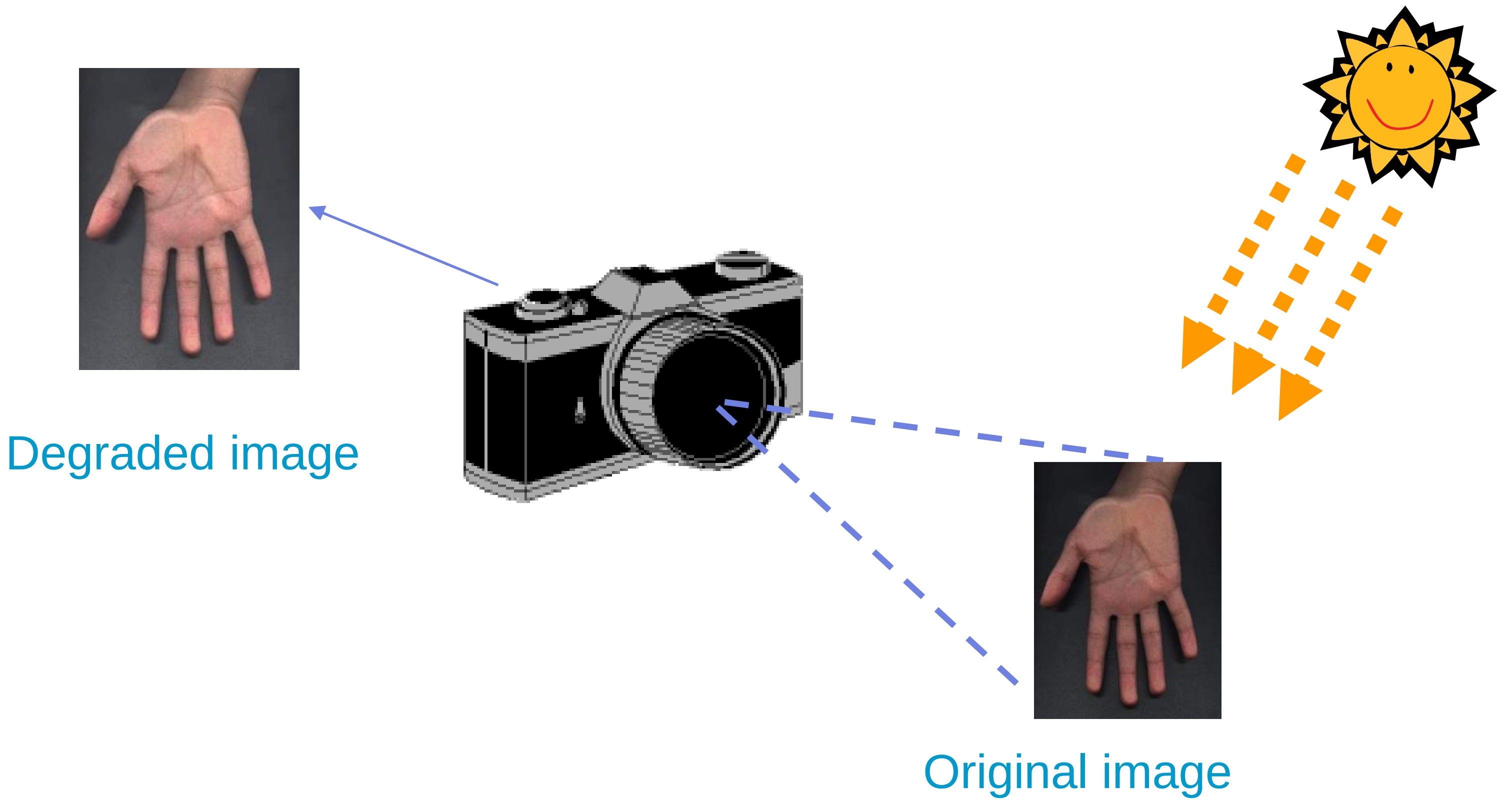


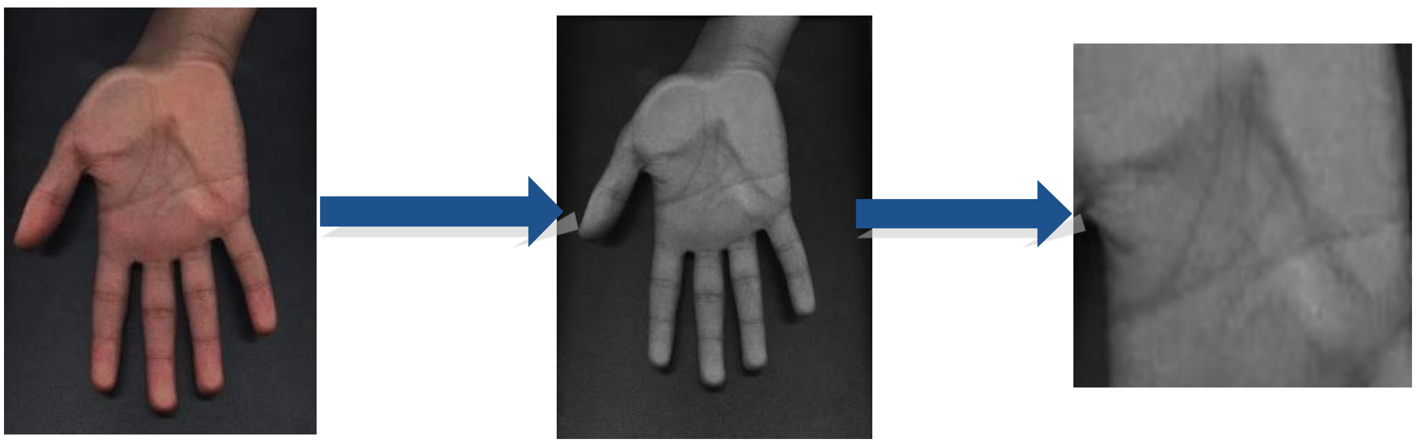
Image Acquisition

A scanner with high resolution



Preprocessing

- Transforming image from RGB to Gray
- Cut only the palm from the hand



Feature Extraction

- EignPalm-based approach to extract the features of palmprint.

- Find the eign-vectors that best account for the distribution of the palmprint image.

- Eignvectors of the covariance matrix palmprint like in appearance, we refer to them as “EignPalms”.

Feature Extraction

- Mathematical Calculations:
- Mean of training palmprints

$$\Psi = \frac{1}{M}\sum_{n=1}^{M}\Gamma_n$$

- Covariance matrix

$$C = \frac{1}{M}\sum_{n=1}^{M}\Phi_n\Phi_n{}^T$$

- N^2 Eigenvectors and eigenvalues

PalmPrint Matching

- The Euclidian distance

$$\varepsilon_k^2 = \| \Omega - \Omega_k \|^2$$

- Chosen threshold (Experimentally) Θ_ε:
- Below Θ_ε : palmprint ‘classified’
- otherwise :palmprint ‘unknown’

Experiment and Results

- Steps:

1- a set of palm images of known persons (5 images for each persons).

2- following the stages as in previous flow chart (acquisition + pre-processing+ feature extraction)

3- using the matlab program developed of algebraic equations (EigenPalm method)

4- Testing

Testing The Program

- Demo Show
- Comments:
- Threshold of 0.75 experminetlly
- Recognition rate up to 90%
- Good rate in recognition world !!

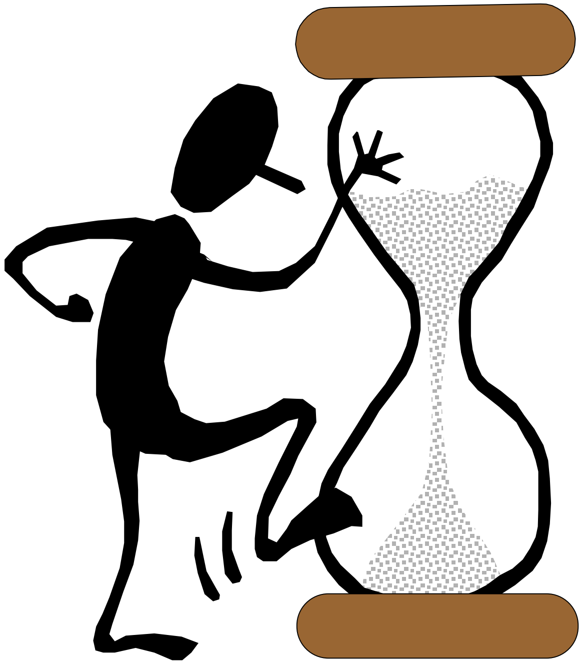
Commercial Application

- Palmprint Identification System

  (Polytechnic University)

Conclusion

- Biometrics system
- Different between PalmPrint &Hand
- PalmPrint Recognition Flow Chart
- Feature Extraction
- Experiment & Results
- Commercial Application



Thanks